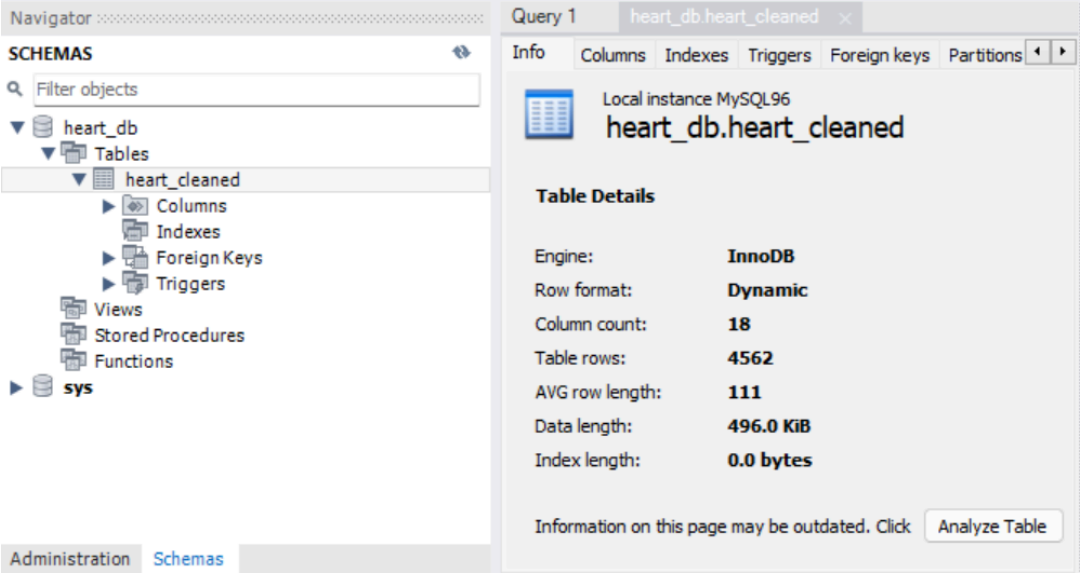
Amount of Data to DB

Heart Disease Analysis Project

Objective

To monitor the volume of data being pulled and rendered from the database and ensure that queries are optimized to prevent system overload.

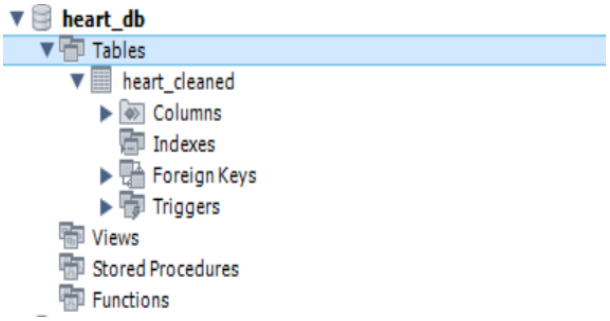
Database Setup

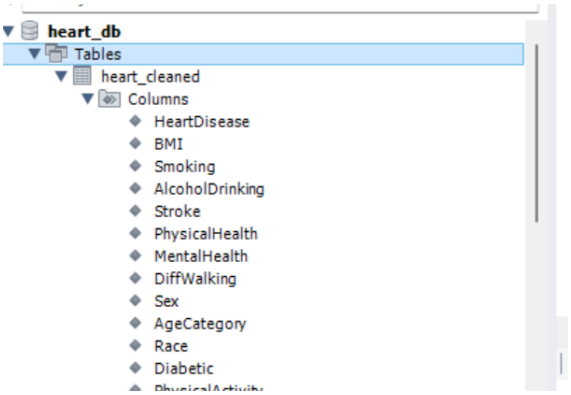


The dataset was initially available in CSV format.
A new database named `heart_db` was created in MySQL Workbench and the CSV file was imported using the *Table Data Import Wizard*.

Table Information

- **Database Name:** heart_db
- **Table Name:** heart_data
- **Total Rows:** 4442
- **Total Columns:** 18
- **Storage Engine:** InnoDB
- **Approx Data Size:** 528 KB





(Table information was verified using the information panel and SQL queries.)

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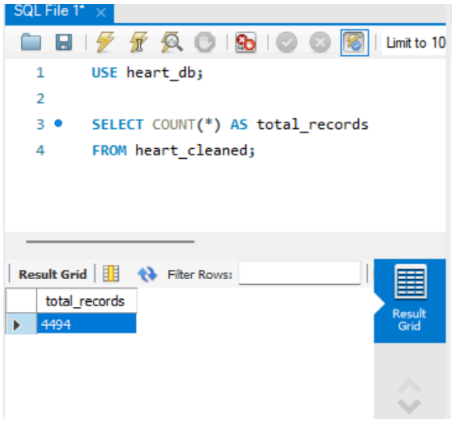
4 SQL Queries Used

<u>To check total number of records:</u>

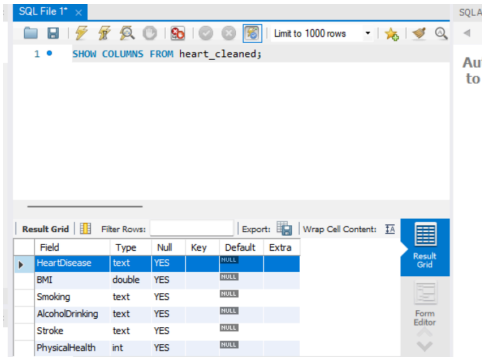
USE heart_db;

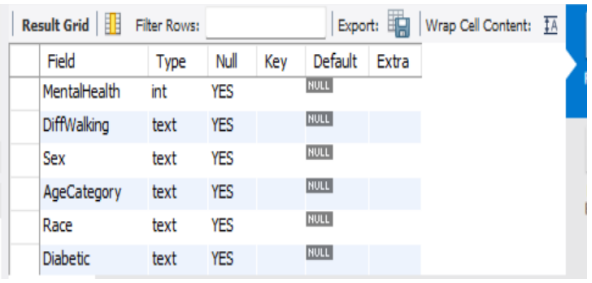
SELECT COUNT(*) AS total_records

FROM heart_cleaned;



To check table structure:





Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
PhysicalActivity	text	YES		NULL	
GenHealth	text	YES		NULL	
SleepTime	int	YES		NULL	
Asthma	text	YES		NULL	
KidneyDisease	text	YES		NULL	
SkinCancer	text	YES		NULL	

```
SHOW COLUMNS FROM heart_cleaned;
```

Performance Monitoring & Optimization

To ensure the system is not overloaded:

- Used specific column selection instead of `SELECT *`
- Applied `WHERE` clause for filtering
- Used `LIMIT` to restrict data output
- Avoided unnecessary joins
- Verified manageable dataset size

Limit to 1000 rows

```
USE heart_db;

SELECT AgeCategory, Sex, BMI
FROM heart_cleaned
WHERE BMI > 25
LIMIT 50;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AgeCategory	Sex	BMI
65-69	Male	26.58
75-79	Female	28.87
80 or older	Female	31.64
80 or older	Female	26.45
65-69	Male	40.69
60-64	Male	34.3
55-59	Female	28.71
75-79	Male	28.37
80 or older	Female	28.15

Result Grid | Filter Rows:

AgeCategory	Sex	BMI
60-64	Male	34.3
55-59	Female	28.71
75-79	Male	28.37
80 or older	Female	28.15
60-64	Female	29.29
50-54	Female	29.18
70-74	Female	26.26
75-79	Female	29.86
70-74	Female	28.9

AgeCategory	Sex	BMI
70-74	Male	29.55
80 or older	Male	27.2
70-74	Female	28.94
75-79	Male	31.46
65-69	Female	26.63
80 or older	Female	34.01
60-64	Male	40
80 or older	Female	27.76
50-54	Male	33.75

optimized query:

```
USE heart_db;
SELECT AgeCategory, Sex, BMI
FROM heart_cleaned
WHERE BMI > 25
LIMIT 50;
```

Observation

The dataset size is moderate and manageable.
The volume of data rendered is controlled through query optimization techniques.
No performance issues were observed during data retrieval.

Conclusion

The amount of data stored in the database has been successfully analyzed.
Proper filtering and optimized queries ensure efficient data handling without overloading the system.